

Aman Kumar

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EDUCATION

IK Gujral Punjab Technical University, Jalandhar

Expected May 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.43 / 10.00

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, DBMS, Computer Networks, Machine Learning

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js
- **Databases & Tools:** MySQL, MongoDB, Git & Github, Postman

WORK EXPERIENCE

Open Source Contributor

June 2025 – Nov 2025

GirlScript Summer of Code

New Delhi, India

- Optimized 10+ HTML templates in a production codebase, improving page structure, load performance, and cross-browser consistency.
- Reviewed and merged 3 GitHub PRs across distributed teams, contributing to technical documentation and scalable code standards.

Full-Stack MERN Trainee

Jun 2025 – July 2025

Excellence Technology

Chandigarh, Punjab, India

- Engineered a full-stack MERN application with JWT auth, real-time chat (Socket.io), and P2P video streaming (WebRTC) cutting API response time by 15% via MongoDB query optimization.
- Designed and deployed 15+ RESTful endpoints in Express.js following MVC architecture, supporting concurrent users with low-latency client-server communication.

RESEARCH

Research Intern – Vicharanashala Lab

May 2026 – Present

Indian Institute of Technology, Ropar

RupNagar, Punjab, India

- Selected for a competitive research residency focusing on open-source infrastructure and full-stack development within the Vicharanashala portfolio.
- Collaborating on the architecture of scalable, user-centric solutions for AI-driven projects like Annam.AI (Agri-tech) and ViBe (Edu-tech).
- Investigating modular system performance and cross-functional technology integration to support high-availability open-source ecosystems.

PROJECTS

PTU Bazar | Multi-Tenant E-Commerce Platform (Ongoing)

GitHub

- Engineered a scalable multi-tenant architecture using Next.js and Node.js to support 50+ independent university vendors and local merchants.
- Designed secure tenant-specific dashboards and authentication modules, reducing merchant onboarding time by 40% through automated store setup.
- Integrated secure payment gateways and real-time tracking, achieving 99.9% system availability via cloud-based deployment and high-availability architecture.

Full Stack Blog Platform

GitHub

- Developed a MERN-based multi-user platform, optimizing API response times by 30% through efficient MongoDB indexing and RESTful design.
- Implemented JWT-based authentication and protected routing, successfully managing 1,000+ secure user sessions with zero reported security breaches.
- Built responsive frontend interfaces using React.js and Material UI, increasing user engagement by 25% across mobile and desktop devices.

LeetLogic | Interview Platform

Demo

- Developed a DSA practice platform aggregating 50+ company-specific interview problems with structured categorization.
- Automated data retrieval and preprocessing from external APIs to analyze frequently occurring coding patterns.

ACHIEVEMENTS

- **Technical & Academic Excellence:** Ranked in the top 10% of the university batch and solved 300+ DSA problems on LeetCode/GFG, earning a 25% University Scholarship for sustained academic performance.
- **Hackathon Recognition:** Secured a Top 10 position out of 500+ competing teams in the Quantum 3.0 Hackathon by developing a platform designed to bridge the skill gap between industry and academia.